Karim Abdel Sadek

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EDUCATION

University of Amsterdam **Amsterdam, NL**
MSc in Artificial Intelligence *Sep 2023 - Present*
Relevant Coursework: Machine Learning I, Deep Learning I&II, Game Theory, Computational Social Choice
GPA: 8.9 / 10 (Equivalent to 4.0/4.0)

Georgia Institute of Technology **Atlanta, USA**
BSc in Computer Science – Exchange student *Jan 2023 – May 2023*
Selected by academic merit to participate in the Exchange Program 2022-2023 with a full ride scholarship.
Coursework: Deep Learning, Information Theory, Quantum Mechanics, Automata and Complexity
GPA: 4.0/4.0

Bocconi University **Milan, IT**
BSc in Mathematics and Computing Sciences for Artificial Intelligence *Sep 2020 – July 2023*
Thesis: '*Machine Learning Augmented Algorithms with Reduced Number of Predictions*', grade: 4/4
Relevant Courses: Real Analysis I,II,III, Game Theory, Physics I,II, Advanced Optimization, Decision Theory
Head Student, Course Representative, Student-Faculty Joint Committees Representative
GPA: 27.8 / 30 (Equivalent to 3.8/4.0)

RESEARCH EXPERIENCE

University of California, Berkeley – Center for Human Compatible AI (CHAI) **Berkeley, US**
Visiting Student Researcher *Jan 2025 – Aug 2025*
Working on Human-AI alignment, social choice theory, multi-agent systems, preference learning
Supervisors: Micah D. Carroll, Anand Siththaranjan, Prof. Stuart Russell

University of Cambridge - Krueger AI Safety Lab **Cambridge, UK**
Research Intern *Mar 2024 – Present*
Working on RL, Goal Misgeneralization, Unsupervised Environment Design
ERA Fellowship Recipient.
Supervisors: Michael D. Dennis, Usman Anwar, David Krueger

Bocconi University, Department of Computing Sciences **Milan, IT**
Research Intern *Jun 2022 – Aug 2023*
First co-author on a paper titled 'Algorithms for Caching and MTS with a reduced number of predictions'
Research Area: Theory of Algorithms, ML-augmented algorithms, Online learning, TCS
Paper has been accepted at ICLR 2024. Top 2% of scores
Supervisor: Marek Elias

Clearbox.AI **Turin, IT**
Machine Learning Intern *Jun 2022 – Aug 2022*
Worked on the generation of synthetic data for internal and business purpose. Benchmark of the SDV MIT library against the internal privacy metrics. Synthesized tens of dataset using both GANs and VAEs.

Bocconi University **Milan, IT**
Teaching Assistant *Feb 2022 – Jan 2023*
TA for the courses 'Algorithms and Computing Theory' (Prof. Luca Trevisan), Physics I,II (Prof. Riccardo Zecchina)

PUBLICATIONS

Karim Abdel Sadek, Marek Elias, "*Algorithms for Caching and MTS with reduced number of predictions*"; *International Conference on Learning Representations (*ICLR*) 2024.* Paper and Reviews

Karim Abdel Sadek*, Matthew F.Roberts***,** Hannah Erlebach, Christian Schroeder De Witt, David Krueger, Usman Anwar, Michael D. Dennis; "*Mitigating goal misgeneralization via minimax regret*", Preprint under Review

Vincenti J*., **Karim Abdel Sadek***, Nulli, M*., Velja, J*., Jazbec M.; "*Dynamic Vocabulary Pruning in Early-Exit LLMs*". ENLSP Workshop, NeurIPS 2024; Paper

Karim Abdel Sadek, Nulli, M., Velja, J., Vincenti J.; "*Explaining RL Decisions with Trajectories: A Reproducibility Study*". Transactions on Machine Learning Research (TMLR) 2024 *; Paper*

SCHOLARSHIPS, AWARDS, and SERVICE

Scholarships and Grants

Cooperative AI PhD Fellowship: Awarded full support for the entire duration of my PhD. It includes the full stipend, in addition to tuition fees, travels, compute, and additional benefits	2025
Berkeley Existential Risk Initiative - Support for research at CHAI, UC Berkeley - $40,000	2024
Krueger AI Safety Lab and ERA Fellowship: Research Intern ’24, ~ $18,000.	2024
Cooperative AI Summer School: Scholarship and Travel Grant, ~ $1500	2024
University of Toronto and CIFAR: AI Inclusive Scholarship for DLRL, ~ $2900 (not used)	2024
Qualcomm Technologies and Ellis Amsterdam: MSc Grant, €2000	2024
University of Amsterdam: MSc Student Travel Grant for ICLR 2024, €500	2024
Inclusive AI Grant, University of Amsterdam, €300	2024
Georgia Institute of Technology, full-ride scholarship for exchange semester Spring ’23, ~$17000	2023
Banca di Credito Cooperativo di Brescia: Merit Scholarship for Academic Achievements, €600	2023
Bocconi University: Funding to present the AILAB winning project at Oxford University, €800	2022
Banca di Credito Cooperativo di Brescia: Merit Scholarship for Academic Achievements, €500	2020
AFS Intercultura: AFS Scolarship for Academic Excellence. Covered the cost of my exchange year in Brazil for the academic 2018/2019, ~€4000	2018
Banca di Credito Cooperativo di Brescia: Merit Scholarship for Academic Achievements, €300	2014

Professional Activities - Reviewing

NeurIPS 2024
ICLR 2025
TMLR

LANGUAGES & SKILLS

Languages: Italian: Native | English: Advanced, IELTS 8.0 | Portuguese: Advanced, C1 | Spanish: Good, Dele B1

Skills: Proficiency in Python | Tensorflow, PyTorch, Jax | R | C

EXTRACURRICULARS,

University of Oxford, Bioinformatics Hub Conference Series

- Winner of the AI LAB competition held by Prof. Francesca Buffa regarding breast cancer detection
- Received funding to travel and present our work at the University of Oxford

Lead The Future Mentorship – *Mentee*

- Among the few Italian students selected to be mentees for Lead The Future, a mentorship non-profit
- Directed to students in STEM, with an acceptance rate below 20%

Student Associations:

- Black in AI
- Inclusive AI, University of Amsterdam
- Head of Research at Bocconi Students for Data Science and AI
- Member at Bocconi Students for Space Developments

Volunteering:

- Teacher Volunteer: Taught math and physics to final year high-school students in a difficult context in Brazil
- AFS Volunteers: I helped and mentored people who will spend a school period abroad to develop the social and cultural skills, and sensibility needed for such life experience. Prepared over 60 students.